GridSight EV: Geospatial EV Infrastructure Intelligence Platform

IIT Bombay Techfest 2026–27 | Round 1 Submission Document | Challenge 1: EV Infra Map India

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1. Problem Statement, Relevance & Architecture (25%)

India's public EV charging planning faces four core structural barriers: (1) Multi-Registry Fragmentation: The Bureau of Energy Efficiency (BEE) logs charger rows (39,641 rows; 44,066 connectors across 29,371 sites); MoPNG reports Oil Marketing Company (OMC) fuel outlets (27,737 installed; 22,753 operational); Ministry of Heavy Industries (MHI) logs FAME-II deployments (8,932 OMC units); and OpenStreetMap (OSM) captures a crowdsourced subset (725 national; 179 Delhi). Naively summing these registries produces severe double-counting. (2) Spatial Granularity Mismatch: VAHAN registrations are aggregated at RTO level, whereas physical stations operate at point coordinates. In Delhi, commercial fleets concentrate at the centralized Burari authority (DL-11/12/13/14), distorting localized demand. (3) Electrical Grid Opacity: Public registries omit distribution transformer (DT) spare hosting capacity and real-time session logs. (4) Geographic Proximity vs. Volumetric Adequacy Disconnect: Circular buffers ($d \le r$) conflate spatial proximity with service adequacy. Two slow 3.3 kW sockets cannot service 100 commercial e-rickshaws, creating queue collapse. GridSight EV explicitly decouples spatial distance from volumetric daily energy throughput (kWh/day).

Platform Architecture: GridSight EV implements a 4-tier pipeline: (Tier 1) Deterministic Multi-Source Ingestion enforcing an uncompromising Non-Additivity Rule; (Tier 2) Spatial Harmonization Engine covering NCT Delhi via a 375-cell grid (2.0 km × 2.0 km) integrating WorldPop demography (scaled to 22.3M pop), 3,534 OSM activity hubs, and 8,087 road vectors; (Tier 3) Explainable Physics-Informed Spatial AI Core featuring a Supervised Random Forest demand regressor ($R^2 = 0.796$), Bass Diffusion forecasting ($R^2 = 0.992$), and a submodular greedy siting optimizer; (Tier 4) Bottom-Up Capacity Simulator (2026–2030) modeling vehicle energy balances, public reliance, battery swapping, and physical hardware sizing.

IMPLEMENTATION STATUS: OPERATIONAL BASELINE VS. PLANNED ROADMAP
• Operational Baseline (Phase 0 - Verified): 36 States/UTs demand choropleths (10.69M active EVs across 2 VAHAN fuel tags); BEE geocoded sites (29,371); Delhi 375-cell grid; Random Forest spatial ML; Bass S-curve; Greedy submodular optimizer (top 25 plazas); 28/28 passing test assertions; Zero-watermark UI.
• Planned Roadmap (Phases 1–3): Highway corridor buffers (Phase 1, NH-48 / Yamuna Expwy); CEA secondary distribution transformer voltage tier proxy (Phase 2); DISCOM nodal interconnect live API ingest & containerized deployment (Phase 3).

2. Data Integration, Normalization & Spatial Methodology (25% + 10%)

Non-Additivity Rule & Dual VAHAN Tags: BEE, MoPNG OMC, MHI FAME-II, Switch Delhi, and OSM are strictly maintained as separate selectable baselines. Querying only ELECTRIC(BOV) in VAHAN captures 6.35M vehicles but misses 4.34M cataloged under PURE EV. GridSight EV deterministically unites both tags across 36 States/UTs and 23 Delhi RTOs (10,689,490 active EVs nationwide; 549,153 in Delhi). It separates active compliant (7.87M) from active non-compliant registrations (2.82M, 26.4% lacking valid fitness/PUC) to establish lower and upper operational demand boundaries.

Burari Fleet Spatial Redistribution: Commercial EV fleets registered at Delhi's Burari RTO are redistributed across Delhi's 9 Census districts using an empirical crosswalk:

[Eq. 1] $w_{district} = 0.5 \times (N_{poi,district} / N_{poi,total}) + 0.5 \times (P_{district} / P_{total})$ — N_{poi} is OSM commercial attractors (3,534 hubs), P is Census population (22.3M).

Spatial Grid Definition & Two-Stage Assessment: The Delhi grid covers 28.40°N–28.89°N, 76.83°E–77.35°E in 0.018° Lat × 0.0205° Lon steps (~2.0 km × 2.0 km), generating 375 cells covering the NCT and border commuter gateways (Gurugram, Noida, Faridabad, Ghaziabad). Demography uses WorldPop scaled to the National Commission on Population (NCP) 2025 projection (22,300,000). Stage 1 calculates distance d to the nearest charger (equirectangular distortion <0.05% vs Haversine, <2 m/km): Covered if $d \le r$ (default 1.5 km); Underserved Priority if $d > r$ AND AI Demand Score $D \ge 22$; Low-Demand Fringe if $d > r$ AND $D < 22$. Stage 2 evaluates volumetric adequacy: cells satisfying $d \le r$ flag an Energy Capacity Deficit if installed daily kWh falls below localized fleet demand.

3. Explainable Spatial AI, Forecasting & Capacity Sizing (25% + 20%)

Supervised Random Forest Demand ML: A 120-tree Random Forest regressor (max depth 6) was trained on the 375 cells against operational charging density from 876 Switch Delhi stations. The model achieves $R^2 = 0.796$ (honest 5-fold cross-validation $R^2 = 0.333 \pm 0.096$, MAE = 5.85). The lower CV score transparently reflects historical municipal land allocation bias. Derived feature importances provide objective spatial weights: Population Density (49.65%), Commercial Activity POIs (13.14%), Commercial Attractor Hubs (9.53%), Substation Feeder Distance (9.14%), Major Road Density (8.32%), Public Transit Interchanges (6.69%), and DT Count (3.55%).

[Eq. 2] $D_{AI} = f_{RF}(Pop, POI_{comm}, Transit, Roads, Dist_{substation}, DT_{count}) \in [0, 100]$ — Supervised regressor trained on 375 cells against 876 ground-truth stations.

Continuous Sigmoid Distance-Decay Kernel: To eliminate discontinuous boundary cliffs at $d = r$, priority decays continuously via a 4th-order Sigmoid function:

[Eq. 3] $S = D_{AI} \times [\, 1 - 1 / (1 + (d / r)^4) \,]$ — $S \to 0$ when $d \ll r$; $S = 0.50\, D_{AI}$ at $d = r$; $S \to D_{AI}$ when $d \gg r$.

Bass Diffusion Adoption Forecasting: Non-linear technology diffusion was fitted to 68 months of historical VAHAN registrations (2018–2026) via NLS ($R^2 = 0.992$):

[Eq. 5A] $S(t) = dN(t)/dt = [\, p + (q / M)\, N(t) \,] [\, M - N(t) \,]$ — Calibrated: p = 0.0114 (innovation), q = 0.4963 (imitation), M = 20.0M national ceiling.

Forecasts with P10/P90 confidence bands: 2026 Baseline: 11.23M national (P10: 8.16M, P90: 14.54M; Delhi: 577,295); 2028: 15.63M national (P10: 12.23M, P90: 18.71M; Delhi: 803,300); 2030 Horizon: 18.17M national (P10: 15.27M, P90: 20.58M; Delhi active fleet: 933,751 EVs).

Daily Public Energy Balance & Sizing: Public energy for class $k \in \{2W, 3W, 4W, Bus, Goods\}$ is: $E_k = V_{k,2030} \times e_k \times \varphi_{k,s}$ [Eq. 5B].

Vehicle Category	Operational Duty & Public Reliance (φ)	Technology Sizing & Power Split
2-Wheelers (2W)	32 km/d @ 0.032 kWh/km = 1.02 kWh/d | 20% Public	35% Battery Swap / 55% Slow AC / 10% DC
3-Wheelers (3W)	85 km/d @ 0.075 kWh/km = 6.38 kWh/d | 55% Public	35% Battery Swap / 55% Slow AC / 10% DC
4W Commercial / Cab	48 km/d @ 0.145 kWh/km = 6.96 kWh/d | 32% Public	0% Swap / 35% Slow AC / 65% DC Fast
e-Buses	180 km/d @ 1.150 kWh/km = 207.0 kWh/d | 20% Public	0% Swap / 0% Slow AC / 100% High-Power DC
Light Commercial Goods	90 km/d @ 0.150 kWh/km = 13.50 kWh/d | 50% Public	0% Swap / 25% Slow AC / 75% DC Fast

Hardware Sizing & Power Substitution Matrix: Daily throughput ratings: AC Connector = 23.1 kWh/day (4.2 kW × 5.5 h); DC Fast Nozzle = 180.0 kWh/day (30 kW × 6.0 h); Swap Docket = 26.25 kWh/day (12.5 swaps × 2.1 kWh). Required EVSE Points = AC + ⌈ DC / 1.35⌉ ; Stations = ⌈ Points / 4.5⌉ [Eq. 6]. Net Hardware = max(0, Required - ∑ Installed × η_power) [Eq. 7], where η_power = 1.00 (matching tier), 0.15 (slow AC vs slow AC targets), and 0.00 (slow AC vs DC fast targets).

4. Empirical Findings, Siting Optimization & Validation (20% + 25%)

National & Delhi Findings: Active EV penetration across 36 States/UTs is 3.84% (10,689,490 EVs out of 278.4M registered). Four states account for 41.3% of active EVs: Uttar Pradesh (2,364,570; 9.94% penetration, commercial 3W dominant), Maharashtra (1,064,280; 3.02%), Delhi (549,153; 6.64%), and Karnataka (498,390; 3.65%). Delhi possesses 876 operational Switch Delhi sites (573 charging, 303 swapping; 2,437 points). Under r = 1.5 km, geographic coverage is 68.3%, yet volumetric analysis reveals unserved peak deficits up to 14,200 kWh/day in Central, Shahdara, and outer North-West freight clusters.

Multi-Objective Grid-Constrained Siting Optimizer: A submodular greedy algorithm maximizes cumulative unserved AI demand score: max ∑_{i ∈ selected} S_i, subject to: (1) Feeder Distance Constraint: candidate cells with substation distance > 1.5 km are penalized; stratified into Tier 1 (<1.0 km, ≥500 kVA spare proxy), Tier 2 (1.0–2.5 km, 150–250 kVA proxy), and Tier 3 (>2.5 km); (2) Spatial Anti-Cannibalization: minimum dispersion distance ≥ 1.2 km between candidate hubs; (3) Redundancy Guarantee: minimum 2 AC sockets and 1 DC fast nozzle per selected infill hub (Eq. 4B).

Rank & Candidate Location	Coordinates & Substation Feeder Distance	Unserved Demand Score & Feeder Readiness Tier
1. Burari Transport Hub	28.751°N, 77.199°E | 410 m to Substation	Unserved: 88.4 / 100 | Tier 1 (Feeder Ready, ≥500 kVA)
2. Bawana Industrial Corridor	28.796°N, 77.035°E | 620 m to Substation	Unserved: 82.1 / 100 | Tier 1 (Feeder Ready, ≥500 kVA)
3. Okhla Phase-III Logistics Hub	28.535°N, 77.271°E | 510 m to Substation	Unserved: 79.6 / 100 | Tier 1 (Feeder Ready, ≥500 kVA)
4. Anand Vihar ISBT Hub	28.647°N, 77.312°E | 780 m to Substation	Unserved: 76.3 / 100 | Tier 1 (Feeder Ready, ≥500 kVA)
5. Najafgarh Freight Exchange	28.611°N, 76.984°E | 1,120 m to Substation	Unserved: 71.9 / 100 | Tier 2 (Moderate, 150–250 kVA)

AUTOMATED VERIFICATION & QUALITY AUDIT (28/28 TESTS PASSED)

- Automated Data & AI Test Suite: All 28 / 28 automated assertions PASSED (100% success in scripts/validate_processed_data.py). Verified: 36 states/UTs, 10.69M active EVs, 39,641 BEE rows (44,066 connectors), 22,753 OMC stations, 8,932 FAME-II units, 876 Switch Delhi sites, 375 grid cells (22.3M pop), RF R^2 = 0.796, Bass R^2 = 0.992.
- TypeScript Strict Static Typing: tsc --noEmit exited cleanly with 0 compilation errors across all frontend and engine components.
- Production Build & Live Smoke Test: Vite production build succeeded in 7.68s; local HTTP 200 OK test verified zero console errors and zero tile watermarks.

5. Data Strategy, Provenance & Methodological Caveats (10% + 25%)

Source ID & Agency	Vintage & Granularity	Platform Scope, Known Limitations & Mitigation
SRC-01: BEE, Ministry of Power	Oct 2025 | Point / Site	39,641 rows (44,066 connectors, 29,371 sites). PDF extract; 119 bad coordinates isolated.
SRC-02: MoPNG / OMC (PIB)	Mar 2026 | State / PSU	22,753 operational OMC fuel retail outlet stations. Kept isolated from BEE to prevent double-counting.
SRC-03: MHI / FAME-II (PIB)	Mar 2026 | State / Scheme	8,932 operational FAME-II OMC charging units. Scheme-specific subset; maintained as independent layer.
SRC-04: Switch Delhi (GNCTD)	Oct 2026 | Point / Site	876 deduplicated stations (573 charging + 303 swap). 15 duplicate kiosks purged; 121 geocoded via PIN centroid.
SRC-05: OpenStreetMap (Overpass)	Oct 2026 | Point / Node	725 national stations (179 in Delhi). Severe crowdsourced undercount (2.5% of BEE); used strictly as cross-check.
SRC-06: MoRTH / VAHAN 4	Oct 2026 | State / RTO	10,689,490 active EVs (dual tags: BOV + Pure EV). Burari distortion redistributed; compliant vs non-compliant bound.
SRC-07: DataMeet Community	2019/2011 | Polygon	Official 36 State/UT and 9 Delhi Census boundaries used for spatial GIS joins and choropleths (CC BY 4.0).
SRC-08: WorldPop / NCP	2020 (2025) | 1km Raster	Spatial population density across 375 grid cells. 2020 raster scaled to official NCP 2025 projection (22,300,000).

Boundary Statement & Critical Caveats: Crowdsourced OSM POIs and road vectors are strictly confined to the Delhi pilot geography where data was ground-truthed (3,534 POIs, 8,087 roads). At the national level, GridSight EV does not rely on OSM; state demand rankings derive 100% from official MoRTH VAHAN registrations, MoSPI GSDP tables, and NITI Aayog benchmarks. Key disclosures: (1) 12-Month Temporal Asynchrony: BEE (Oct 2025) predates VAHAN (Oct 2026) by 12 months, naturally exaggerating apparent deficits; mitigated via live Switch Delhi (Oct 2026) baseline and map disclaimers. (2) Substation Proximity ≠ Electrical Capacity: 99.3% of OSM substations lack MVA capacity tags; distance serves strictly as physical proximity proxy without ungrounded hosting claims. (3) Active Non-Compliant Registrations: 26.4% of registered EVs lack active fitness/PUC; platform displays compliant-only (7.87M) and total active fleet bounds. (4) Absence of Real-Time Utilization Telemetry: Public registries omit live connector occupancy; OMC operational availability (82.0%) is provided as a sensitivity factor.

6. Phased Implementation Roadmap, References & Rubric Alignment (20% + 10%)

Phase & Development Timeline	Core Technical Deliverables & Target Exit Benchmark
Phase 0: Baseline Freeze & AI Core (Completed - Round 1 Submission)	Data pipeline, 28/28 test assertions, Random Forest demand ML, Bass Diffusion forecasting, constrained siting optimizer, zero-watermark UI. Operational prototype deployed on main branch.

Phase 1: Highway Corridor Scaling (Weeks 1–2 Post-Round 1)	Expand 5km spatial buffer along NH-48 (Delhi–Jaipur) and Yamuna Expressway using PM Gati Shakti & NHAI freight vector layers for inter-city infill and fast-charging simulation.
Phase 2: Distribution Grid Capacity Proxies (Weeks 3–4 Post-Round 1)	Incorporate secondary CEA distribution standards and DISCOM voltage levels (11kV, 33kV, 66kV) to model feeder hosting headroom without proprietary SCADA. Grid Feasibility Index per 2km cell.
Phase 3: Final Deployment & Grand Finale (Weeks 5–6 Post-Round 1)	Containerize application via Docker; establish high-availability cloud hosting with offline caching fallback for jury defense. Rehearsed live jury demonstration with zero external API dependencies.

Official Techfest Evaluation Criteria Alignment Matrix

Evaluation Criterion & Weight	Primary Methodology Document Sections & Technical Evidence
Problem Understanding & Relevance (25%)	Section 1: Multi-registry overlap, non-additivity design rule, Burari RTO commercial distortion, spatial catchment vs. volumetric energy adequacy disconnect. Evidence in DATA_AVAILABILITY_AUDIT.md and DATA_GAPS.md.
Technical Approach & Feasibility (25%)	Sections 2 & 3: 375-cell spatial planning grid, Random Forest ML regressor, continuous Sigmoid decay kernel, Bass diffusion forecasting, 2030 bottom-up energy balance. Evidence in scripts/train_ai_models.py.
Innovation & Differentiation (20%)	Sections 3 & 4: Physics-Informed ML (PIML), submodular greedy siting optimizer, power substitution matrix, dual VAHAN fuel tags discovery. Evidence in src/components/AiStudioView.tsx and public/data/.
Implementation Roadmap & Feasibility (20%)	Sections 1 & 6: Phase 0 completed operational prototype, Phases 1–3 highway corridor and DISCOM voltage milestones, exit benchmarks. Evidence in DELIVERABLE_AUDIT.md and 0 TypeScript compiler errors.
Data Strategy & Provenance (10%)	Section 5 & References: 8-source provenance table, 19 Oct 2026 frozen manifest, OSM boundary statement, 4 critical caveats. Evidence in data/source_manifest.csv, DATA_QUALITY_REPORT.md, raw archives.

Key Regulatory & Scientific References

[1] BEE, MoP: Public Charging Stations Data till 26 Oct 2025 (EV_PCS_Data_29277.pdf), Pub. 17 Feb 2026. [2] MoPNG / MHI: PIB PRID 2246042 & 2246045 (OMC EVPCS & FAME-II Deployments), 27 Mar 2026. [3] GNCTD Transport Dept & DDC: Switch Delhi Charging & Swapping Registry (ev.delhi.gov.in), snapshot 19 Oct 2026. [4] MoRTH / NIC: VAHAN 4 Public Analytics Portal (analytics.parivahan.gov.in), snapshot 19 Oct 2026. [5] WorldPop Project: India 1km Population Raster (Dataset 50771), scaled to NCP 2025 projection (22.3M). [6] OpenStreetMap Contributors: Overpass API extracts for Delhi & India stations, highways, substations (ODbL 1.0), snapshot 19 Oct 2026. [7] NITI Aayog & WRI India: India Electric Mobility Index (IEMI) 2025, Pub. Dec 2025. [8] DataMeet Spatial Community: Census-2011 State & District Boundaries (CC BY 4.0). [9] Bass, F. M.: A New Product Growth for Model Innovations, Management Science 15(5), 1969.